Subham Ghosh

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GitHub | LinkedIn | Portfolio

EDUCATION

Bachelor of Engineering in Information Science Sept 2023 – Aug 2027
Dr. Ambedkar Institute of Technology, Bangalore, India **CGPA: 8.18/10**

TECHNICAL SKILLS

Languages: Python, JavaScript, C#
Frontend: React.js, HTML5, CSS, Tailwind CSS
Backend: Node.js, Express.js, Flask, RESTful APIs
Databases: SQL, MySQL, MongoDB
Cloud & DevOps: AWS Amplify, Render, Linux, Docker (Fundamentals), CI/CD (Fundamentals)
Tools: Git, GitHub, Postman, VS Code
Core CS: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

EXPERIENCE

MCT (Startup) Dec 2025 – Present

- Developed and maintained the core React.js frontend and application workflows; deployed the production application using AWS Amplify.
- Coordinated development tasks and feature integration across a 3-member team, aligning frontend requirements with backend API contracts.
- Managed Git-based version control, code integration and deployment handoffs to maintain a stable codebase.

PROJECTS

PotCity – Smart City Road Maintenance Simulation *C#, Unity, BFS, State Machines*

- Built a Unity-based simulation automating the road maintenance lifecycle using autonomous survey and repair vehicle agents.
- Implemented BFS-based graph navigation for autonomous vehicles across multi-lane road networks and junctions.
- Designed a decoupled, event-driven architecture and state-machine-driven pothole lifecycle with resource- constrained repair scheduling.
- Engineered a physics-based spatial scanning system to estimate pothole dimensions and calculate repair costs.

MediGuide AI – Kidney Disease Prediction System *Python, Flask, React, MongoDB, TensorFlow*

- Developed a multi-model prediction pipeline combining Random Forest for clinical data, CNN for medical image classification and LSTM for disease progression prediction.
- Built Flask REST APIs to expose model inference and integrated backend services with a React frontend and MongoDB.
- Implemented preprocessing pipelines for clinical and imaging datasets to support model training and inference.

AgriBoost – Smart Agriculture Platform *Node.js, Express.js, EJS, MongoDB*

- Developed a full-stack Progressive Web App providing crop health insights and weather-based decision support for farmers.
- Implemented JWT-based authentication, input validation and error handling; integrated external weather APIs and automated agricultural news aggregation.
- Optimised the application for mobile-first, low-bandwidth usage and deployed production services on Render.

CERTIFICATIONS & ACHIEVEMENTS

- Oracle Cloud Infrastructure (OCI) AI Foundations Associate.
- Finalist – Buildverse National Level Hackathon.
- Participant – Google Solution Challenge.
- Participant – GDG-ML Machine Learning Event.